

SAT Math Mastery · Week 5A

Exponentials & Sequences

Time 150 minutes: 110 min solving + 40 min self-check and error log

Topics exponential growth and decay models · base and initial value · doubling and half-life · compound growth · exponent laws · arithmetic vs geometric sequences · linear vs exponential · tables and graphs

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Build $y = ab^t$ or $y = a(1 \pm r)^{t/k}$ from a verbal description (percent $\rightarrow$ factor; “every k units” $\rightarrow$ exponent t/k).
- S2.** Read the initial value and the growth or decay factor from an equation, a table or a graph.
- S3.** Use doubling time and half-life; solve for a time by matching powers of the same base.
- S4.** Tell exponential from linear: constant *ratio* versus constant *difference*.
- S5.** Apply the exponent laws, rational exponents and equal-base equations; rewrite equivalent exponential forms.
- S6.** Arithmetic sequences $a_n = a_1 + (n-1)d$ and geometric sequences $a_n = a_1r^{n-1}$ (including negative ratios).
- S7.** Interpret constants and factors in context; identify the growth type from “equals $p\%$ of the previous value”.
- S8.** Build a and b from two points; use parameters, shifts and hidden conditions.

2 · REQUIRED FACTS

- Exponential: equal steps in x multiply y by the same factor. Linear: equal steps in x add the same amount to y .
- $y = ab^x$ with $a > 0$: $b > 1$ is growth, $0 < b < 1$ is decay; a is the value at $x = 0$ and the graph never touches the x -axis.
- A change of $p\%$ per period has factor $1 + \frac{p}{100}$ (growth) or $1 - \frac{p}{100}$ (decay), never $\frac{p}{100}$ alone.
- “Every k units” puts $\frac{t}{k}$ in the exponent; the factor for one unit is the k -th root of the factor for k units.
- The exponent in a geometric sequence is $n - 1$, not n . Terms m apart differ by the factor r^m (or by md in an arithmetic sequence).
- A compounded balance over n periods is $P(1 + \frac{r}{n})^{nt}$ for annual rate r , n periods per year.

3 · REQUIRED FORMULAS

$$y = ab^x, \quad y = a(1 \pm r)^t, \quad y = ab^{t/k}, \quad a_n = a_1r^{n-1}, \quad a_n = a_1 + (n-1)d$$

$$b^m b^n = b^{m+n}, \quad \frac{b^m}{b^n} = b^{m-n}, \quad (b^m)^n = b^{mn}, \quad b^{-n} = \frac{1}{b^n}, \quad b^{m/n} = \sqrt[n]{b^m}, \quad b^0 = 1$$

$$\text{two points: } b^{x_2-x_1} = \frac{y_2}{y_1}, \quad A = P \left(1 + \frac{r}{n}\right)^{nt}, \quad \text{half-life: } N = N_0 \left(\frac{1}{2}\right)^{t/h}$$

4 · MINI THEORY SHEET

Linear or exponential?

	Linear	Exponential
Table	equal Δy for equal Δx	equal y_2/y_1 for equal Δx
Words	“increases by c each period”	“increases by $p\%$ each period”
Equation	$y = mx + b$	$y = ab^x$
Long run	grows by the same amount	eventually beats every linear model

Reading $y = ab^t$. a = value when $t = 0$. b = factor per unit of t ; the percent change is $(b - 1) \cdot 100\%$. If the exponent is t/k , then b is the factor per k units.

Rewriting the base. $b^{t/k} = (b^{1/k})^t$. Write both sides over one common base (powers of 2, 3, 5) and compare exponents.

From a table or two points. Divide the outputs and take a root of the step count: if x moves by m , then $b^m = \frac{y_2}{y_1}$. Then $a = y_1 b^{-x_1}$.

Sequences. Same ratio $\Rightarrow$ geometric, same difference $\Rightarrow$ arithmetic. Three consecutive terms p, q, r of a geometric sequence satisfy $q^2 = pr$; of an arithmetic sequence, $2q = p + r$.

Context. The constant out front is the starting amount; the base is the multiplier per period; never read the exponent as the percent.

Speed shortcuts.

Situation	Shortcut
doubling / halving	count how many doublings: 2^n ; target = start $\times 2^n$
percent $\rightarrow$ factor	$+7\% \rightarrow 1.07$, $-12\% \rightarrow 0.88$; multiply by the factor n times
every k units	exponent $\frac{t}{k}$; rewrite as $(b^{1/k})^t$ for a per-unit factor
solve for an expression	factor out 3^x (e.g. $3^{x+2} - 3^x = 8 \cdot 3^x$) and read off the block
two table rows	$b^{\Delta x} = \frac{y_2}{y_1}$, then work back to a with one equation
geometric three-in-a-row	$q^2 = pr$ gives the unknown immediately
linear vs exponential race	test values; the exponential only wins after a crossover month
shift in a graph	$f(x - h) = a b^{-h} b^x$: the shift changes only the leading constant

5 · GUIDED EXAMPLES

Example 1 — Build the model from words

A medicine concentration starts at 240 mg/L and falls by 30% every 4 hours. Write $C(t)$, t in hours, and find $C(8)$.

Factor per 4 hours: $1 - 0.30 = 0.70$. So $C(t) = 240(0.7)^{t/4}$.

$C(8) = 240(0.7)^2 = 240(0.49) = 117.6$ mg/L.

Example 2 — Two points $\rightarrow a$ and b

An exponential function has $f(1) = 12$ and $f(4) = 96$. Find $f(x)$ and $f(0)$.

Three steps from $x = 1$ to $x = 4$: $b^3 = \frac{96}{12} = 8 \Rightarrow b = 2$.

$12 = a(2)^1 \Rightarrow a = 6$. So $f(x) = 6(2)^x$ and $f(0) = 6$.

Example 3 — Match the base

Solve $4^{x+1} = 32^{x-2}$.

$4 = 2^2$ and $32 = 2^5$: $2^{2(x+1)} = 2^{5(x-2)}$. Exponents: $2x + 2 = 5x - 10 \Rightarrow x = 4$.

6 · PRACTICE SET — 48 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 3 \cdot 2^x$. What is the value of $f(5) - f(3)$?

1. $f(5) = 3 \cdot 2^5 = 3 \cdot 32 = 96$.
2. $f(3) = 3 \cdot 2^3 = 3 \cdot 8 = 24$.
3. $f(5) - f(3) = 96 - 24 = 72$.

Answer: 72

2.

EASY · NO CALCULATOR NEEDED

The table shows values of the function f for selected values of x . Which of the following defines f ?

x	0	1	2	3
$f(x)$	64	48	36	27

(A) $f(x) = 64 \left(\frac{1}{4}\right)^x$

(B) $f(x) = 64 \left(\frac{3}{4}\right)^x$ ✓

(C) $f(x) = 64 - 16x$

(D) $f(x) = 64 \left(\frac{4}{3}\right)^x$

1. Each output is the previous one times the same factor: $\frac{48}{64} = \frac{36}{48} = \frac{27}{36} = \frac{3}{4}$.
2. The differences $(-16, -12, -9)$ are not constant, so the function is not linear.
3. With $f(0) = 64$ and ratio $\frac{3}{4}$: $f(x) = 64 \left(\frac{3}{4}\right)^x$.

Answer: (B)

3.

EASY · NO CALCULATOR NEEDED

The balance, in dollars, of a savings account t years after the account is opened is modeled by $A(t) = 800(1.05)^t$. Which of the following statements best describes the account?

(A) Starts at \$800; up 105% per year.

(B) Starts at \$1.05; up \$800 per year.

(C) Starts at \$800; up 5% per year. ✓

(D) Starts at \$800; up \$5 per year.

1. In $A(t) = ab^t$, $a = 800$ is the starting balance and $b = 1.05$ is the yearly factor.
2. A factor of $1.05 = 1 + 0.05$ means the balance grows by 5% each year (not by \$5, and not by 105%).

Answer: (C)

4.

EASY · NO CALCULATOR NEEDED

The first term of a geometric sequence is 6 and the common ratio is $\frac{1}{2}$. What is the fourth term of the sequence?

(A) $\frac{3}{4}$ ✓

(B) $\frac{3}{8}$

(C) $\frac{3}{2}$

(D) $\frac{3}{16}$

1. $a_n = a_1 r^{n-1}$, so $a_4 = 6 \left(\frac{1}{2}\right)^3$.

2. $a_4 = 6 \cdot \frac{1}{8} = \frac{3}{4}$.

Answer: (A)

5.

EASY · NO CALCULATOR NEEDED

Which of the following is equivalent to $\frac{(2x^3)^3}{4x^2}$, for $x \neq 0$?

(A) $\frac{x^7}{2}$

(B) $2x^{11}$

(C) $2x^{\frac{9}{2}}$

(D) $2x^7$ ✓

1. Cube everything inside: $(2x^3)^3 = 2^3 x^9 = 8x^9$.

2. Divide: $\frac{8x^9}{4x^2} = 2x^{9-2} = 2x^7$.

Answer: (D)

6.

MEDIUM · NO CALCULATOR NEEDED

A lab culture contains 500 bacteria at time $t = 0$, where t is measured in hours, and the number of bacteria doubles every 3 hours. Which of the following equations gives the number of bacteria N at time t ?

(A) $N = 500(2)^t$

(B) $N = 500(2)^{3t}$

(C) $N = 500(2)^{t/3}$ ✓

(D) $N = 500(3)^t$

1. Doubling every 3 hours means t hours contain $\frac{t}{3}$ doublings.

2. $N = 500 \cdot 2^{t/3}$. Check at $t = 3$: $500 \cdot 2^1 = 1000$ (doubled once).

3. The other options fail the same check: $500(2)^3 = 4000$, $500(2)^9 = 256,000$ and $500(3)^3 = 13,500$ at $t = 3$.

Answer: (C)

7.

MEDIUM · NO CALCULATOR NEEDED

A 640-milligram sample of a radioactive substance loses half of its mass every 5 days. After how many days will exactly 20 milligrams of the sample remain?

1. Halve repeatedly: $640 \rightarrow 320 \rightarrow 160 \rightarrow 80 \rightarrow 40 \rightarrow 20$.

2. That is 5 halvings, and each takes 5 days.

3. $5 \times 5 = 25$ days. (Equivalently $640 \left(\frac{1}{2}\right)^{t/5} = 20 \Rightarrow 2^{t/5} = 32 = 2^5 \Rightarrow t = 25$.)

Answer: 25

8.

MEDIUM · CALCULATOR

A delivery van is bought for \$28,000 and loses 12% of its value each year. To the nearest dollar, what is the value of the van 3 years after it was purchased?

(A) \$17,920

(B) \$19,081 ✓

(C) \$24,640

(D) \$39,338

1. Each year the value is multiplied by $1 - 0.12 = 0.88$.

2. $V(3) = 28,000(0.88)^3 = 28,000(0.681472) = 19,081.2$.

3. To the nearest dollar: \$19,081. (Subtracting 12% of the original price three times gives \$17,920, which treats the decay as linear.)

Answer: (B)

9.

MEDIUM · NO CALCULATOR NEEDED

The exponential function f satisfies $f(2) = 18$ and $f(5) = 486$. What is the value of $f(0)$?

1. From $x = 2$ to $x = 5$ is 3 steps: $b^3 = \frac{486}{18} = 27 \Rightarrow b = 3$.

2. $f(x) = a(3)^x$ and $f(2) = 18$: $9a = 18 \Rightarrow a = 2$.

3. $f(0) = a = 2$.

Answer: 2

10.

MEDIUM · NO CALCULATOR NEEDED

The first term of a geometric sequence is 5 and the fourth term is 135. What is the second term?

(A) 27

(B) 45

(C) 70

(D) 15 ✓

1. $a_4 = a_1 r^3 \Rightarrow 135 = 5r^3 \Rightarrow r^3 = 27 \Rightarrow r = 3.$

2. $a_2 = a_1 r = 5 \cdot 3 = 15.$

Answer: (D)

11.

MEDIUM · NO CALCULATOR NEEDED

Each year, the population of a town is 94% of its population the previous year. Which of the following types of function best models the town's population as a function of the number of years?

(A) Exponential decay ✓

(B) Increasing linear

(C) Decreasing linear

(D) Exponential growth

1. "Is 94% of the previous value" is a multiplication by the same factor 0.94 each year, so the model is exponential, not linear.

2. The factor $0.94 < 1$, so the population shrinks: exponential decay.

Answer: (A)

12.

MEDIUM · NO CALCULATOR NEEDED

The value, in dollars, of a collectible t years after 2020 is modeled by $V(t) = 1800(0.92)^t$. Which of the following is the best interpretation of the number 0.92 in this context?

(A) The value in 2020 was \$0.92.

(B) The value decreases by 8% each year. ✓

(C) The value decreases by 92% each year.

(D) The value decreases by \$0.92 each year.

1. 0.92 is the factor applied each year: each year's value is 92% of the previous year's.

2. A factor of $0.92 = 1 - 0.08$ is a decrease of 8% per year (not 92%, and not a \$0.92 drop).

Answer: (B)

13.

MEDIUM · NO CALCULATOR NEEDED

If $27^{x-1} = 9^{x+2}$, what is the value of x ?

1. Write both sides with base 3: $27 = 3^3$ and $9 = 3^2$.
2. $3^{3(x-1)} = 3^{2(x+2)} \Rightarrow 3x - 3 = 2x + 4$.
3. $x = 7$. (Check: $27^6 = 3^{18} = 9^9$.)

Answer: 7**14.**

HARD · CALCULATOR

A savings account is modeled by $A(t) = 5000(1.015)^{4t}$, where $A(t)$ is the balance in dollars t years after the initial deposit. Which of the following is closest to the percent by which the balance increases each year?

- (A) 1.50% (B) 6.00%
(C) 6.14% ✓ (D) 15.0%

1. The exponent $4t$ means the factor 1.015 is applied 4 times per year.
2. Yearly factor: $(1.015)^4 \approx 1.0614$.
3. The yearly increase is about 6.14% (not 6%, which ignores compounding, and not 1.5%, which is per quarter).

Answer: (C)**15.**

HARD · NO CALCULATOR NEEDED

The function f is defined by $f(t) = 90 \cdot 8^{t/6}$. Which of the following functions is equivalent to f ?

- (A) $90(\sqrt{2})^t$ ✓ (B) $90 \cdot 2^{t/6}$
(C) $90\left(\frac{4}{3}\right)^t$ (D) $90 \cdot 4^{t/6}$

1. Rewrite $8 = 2^3$: $8^{t/6} = 2^{3t/6} = 2^{t/2}$.
2. $2^{t/2} = \left(2^{1/2}\right)^t = (\sqrt{2})^t$, so $f(t) = 90(\sqrt{2})^t$.
3. Check at $t = 6$: $90 \cdot 8 = 720$ and $90(\sqrt{2})^6 = 90 \cdot 8 = 720$.

Answer: (A)

16.

HARD · CALCULATOR

For the exponential function $f(x) = a \cdot b^x$, where a and b are positive constants, $f(1) = 24$ and $f(4) = 81$. What is the value of $f(6)$?

(A) 243

(B) $\frac{2187}{8}$ (C) $\frac{243}{2}$ (D) $\frac{729}{4}$ ✓

1. Three steps from $x = 1$ to $x = 4$: $b^3 = \frac{81}{24} = \frac{27}{8} \Rightarrow b = \frac{3}{2}$.

2. $24 = a \cdot \frac{3}{2} \Rightarrow a = 16$, so $f(x) = 16 \left(\frac{3}{2}\right)^x$.

3. $f(6) = 16 \cdot \frac{729}{64} = \frac{729}{4} = 182.25$. (Shortcut: $f(6) = f(4) \cdot b^2 = 81 \cdot \frac{9}{4} = \frac{729}{4}$.)

Answer: (D)

17.

HARD · DESMOS

Channel A has 50 subscribers and gains 30 subscribers each month. Channel B has 50 subscribers and its subscriber count increases by 20% each month. What is the least whole number of months after which Channel B has more subscribers than Channel A?

1. A: $50 + 30m$. B: $50(1.2)^m$. Both start at 50.

2. For small m the linear channel leads (e.g. $m = 6$: 230 vs ≈ 149 ; $m = 10$: 350 vs ≈ 309.6 ; $m = 11$: 380 vs ≈ 371.5).

3. $m = 12$: A has 410 and B has $50(1.2)^{12} \approx 445.9$. B first exceeds A at $m = 12$.

Answer: 12

18.

HARD · NO CALCULATOR NEEDED

The second term of a geometric sequence is 6, the fourth term is 54, and the common ratio is negative. What is the fifth term?

1. $a_4 = a_2 r^2 \Rightarrow 54 = 6r^2 \Rightarrow r^2 = 9$.

2. The ratio is negative, so $r = -3$.

3. $a_5 = a_4 \cdot r = 54(-3) = -162$.

Answer: -162

19.

HARD · CALCULATOR

A radioactive sample loses 36% of its mass every 5 days. Its mass can also be modeled by $M(t) = M_0 b^t$, where t is the number of days and b is a constant. What is the value of b , rounded to the nearest thousandth?

1. Every 5 days the mass is multiplied by $1 - 0.36 = 0.64$: $b^5 = 0.64$.

2. $b = 0.64^{1/5} \approx 0.9146$.

3. Rounded to the nearest thousandth: 0.915.

Answer: 0.915

20.

HARD · NO CALCULATOR NEEDED

If $3^{x+2} - 3^x = 216$, what is the value of 3^{x+1} ?

1. Factor out 3^x : $3^{x+2} - 3^x = 3^x(9 - 1) = 8 \cdot 3^x$.

2. $8 \cdot 3^x = 216 \Rightarrow 3^x = 27$.

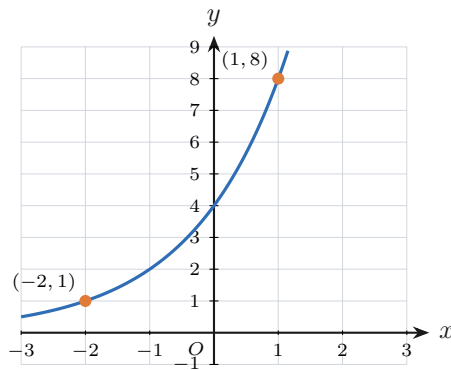
3. $3^{x+1} = 3 \cdot 3^x = 3 \cdot 27 = 81$.

Answer: 81

21.

HARD · DESMOS

The graph of the function $f(x) = a \cdot b^x$, where a and b are positive constants, is shown in the xy -plane. The points $(-2, 1)$ and $(1, 8)$ lie on the graph. What is the value of $a + b$?



(A) 10

(B) 12

(C) 6 ✓

(D) 8

1. Read the two marked points: $f(-2) = 1$ and $f(1) = 8$.
2. Three steps from $x = -2$ to $x = 1$: $b^3 = \frac{8}{1} = 8 \Rightarrow b = 2$.
3. $f(1) = a \cdot 2 = 8 \Rightarrow a = 4$ (check: $4 \cdot 2^{-2} = 1$).
4. $a + b = 4 + 2 = 6$.

Answer: (C)

22.

HARD · NO CALCULATOR NEEDED

The numbers $x + 2$, $x + 8$ and $x + 26$ are, in that order, three consecutive terms of a geometric sequence. What is the value of x ?

(A) 6

(B) 1 ✓

(C) 2

(D) 4

1. In a geometric sequence the middle term squared equals the product of its neighbors: $(x + 8)^2 = (x + 2)(x + 26)$.
2. $x^2 + 16x + 64 = x^2 + 28x + 52 \Rightarrow 12 = 12x \Rightarrow x = 1$.
3. Check: 3, 9, 27 with ratio 3.

Answer: (B)

23.

HARD · NO CALCULATOR NEEDED

For positive x and y , the expression $\frac{(x^{2/3}y^{-1})^3}{x^{-1}y^2}$ is equivalent to x^py^q , where p and q are constants. What is the value of $p + q$?

(A) 0

(B) 8

(C) -8

(D) -2 ✓

1. Numerator: $(x^{2/3}y^{-1})^3 = x^2y^{-3}$.

2. Divide by $x^{-1}y^2$ (subtract exponents): $x^{2-(-1)}y^{-3-2} = x^3y^{-5}$.

3. $p = 3$, $q = -5$, so $p + q = -2$.

Answer: (D)

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

An arithmetic sequence a_n and a geometric sequence g_n both have first term 2. The common difference of a_n and the common ratio of g_n are both equal to k , where $k > 1$. If $g_3 - a_3 = 38$, what is the value of g_5 ?

1. $a_3 = 2 + 2k$ and $g_3 = 2k^2$.

2. $2k^2 - (2 + 2k) = 38 \Rightarrow 2k^2 - 2k - 40 = 0 \Rightarrow k^2 - k - 20 = 0 \Rightarrow (k - 5)(k + 4) = 0$.

3. Since $k > 1$, $k = 5$.

4. $g_5 = 2 \cdot 5^4 = 2 \cdot 625 = 1250$.

Answer: 1250

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

For the function $f(x) = a \cdot b^x$, where a and b are constants and $b > 0$, $f(2) = 12$ and $f(4) - f(2) = 96$. What is the value of $f(5)$?

1. $f(4) = f(2)b^2 = 12b^2$, so $f(4) - f(2) = 12b^2 - 12 = 96$.
2. $b^2 = 9$; since $b > 0$, $b = 3$.
3. $f(5) = f(2)b^3 = 12 \cdot 27 = 324$. (Also $a = \frac{12}{9} = \frac{4}{3}$ and $f(5) = \frac{4}{3} \cdot 243 = 324$.)

Answer: 324**26.**

SAT CHALLENGE · NO CALCULATOR NEEDED

Culture A starts with 4000 cells and its population is cut in half every 2 hours. Culture B starts with 500 cells and its population doubles every 3 hours. After how many hours, t , will the two cultures have the same number of cells?

(A) 3.6 ✓

(B) 18

(C) 0.6

(D) 1.8

1. A: $4000 \left(\frac{1}{2}\right)^{t/2} = 4000 \cdot 2^{-t/2}$. B: $500 \cdot 2^{t/3}$.
2. Set equal and divide by 500: $8 \cdot 2^{-t/2} = 2^{t/3} \Rightarrow 2^{3-t/2} = 2^{t/3}$.
3. $3 - \frac{t}{2} = \frac{t}{3} \Rightarrow 3 = \frac{5t}{6} \Rightarrow t = \frac{18}{5} = 3.6$ hours.

Answer: (A)

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

The terms of a geometric sequence are $a_1, a_2, a_3, \dots$. The sum of the second and third terms is 12, and the sum of the third and fourth terms is 36. What is the value of a_6 ?

1. $a_3 + a_4 = r(a_2 + a_3)$, so $r = \frac{36}{12} = 3$.
2. $a_2 + a_3 = a_2(1 + 3) = 12 \Rightarrow a_2 = 3$, so $a_1 = 1$.
3. $a_6 = a_1 r^5 = 3^5 = 243$.

Answer: 243**28.**

SAT CHALLENGE · NO CALCULATOR NEEDED

A quantity decreases by 19% every 2 years. Its current value is 4000. In how many years from now will its value be 2916?

1. Every 2 years the factor is $1 - 0.19 = 0.81 = 0.9^2$, so the yearly factor is 0.9.
2. $V(t) = 4000(0.9)^t$. Set $4000(0.9)^t = 2916 \Rightarrow 0.9^t = 0.729$.
3. $0.729 = 0.9^3$, so $t = 3$ years. (Check: $4000 \cdot 0.729 = 2916$.)

Answer: 3

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = a \cdot b^x$, where a and b are positive constants. If $f(3) > f(2)$, which of the following statements must be true?

I. $f(-1) < f(0)$ II. $f(5) - f(4) > f(4) - f(3)$ III. $f(0) > 1$

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II and III

1. $f(3) > f(2) \iff ab^3 > ab^2 \iff b > 1$ (since $a, b > 0$). The function is increasing.

2. I: increasing, so $f(-1) < f(0)$. True.

3. II: $f(5) - f(4) = ab^4(b - 1)$ and $f(4) - f(3) = ab^3(b - 1)$; the first is b times the second and $b > 1$, so it is larger. True.

4. III: only $a > 0$ is known; for example $a = \frac{1}{2}$, $b = 2$ gives $f(0) = \frac{1}{2} < 1$. Not necessarily true.

5. Only I and II must be true.

Answer: (B)**30.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 48 \left(\frac{2}{3}\right)^x$. The function g is defined by $g(x) = f(2x - 4)$ and can be written in the form $g(x) = c \cdot d^x$, where c and d are constants. What is the value of $c \cdot d$?

(A) 243

(B) 48

(C) 108 ✓

(D) 162

$$1. g(x) = 48 \left(\frac{2}{3}\right)^{2x-4} = 48 \left(\frac{2}{3}\right)^{-4} \left[\left(\frac{2}{3}\right)^2\right]^x.$$

$$2. d = \left(\frac{2}{3}\right)^2 = \frac{4}{9} \text{ and } c = 48 \left(\frac{3}{2}\right)^4 = 48 \cdot \frac{81}{16} = 243.$$

$$3. c \cdot d = 243 \cdot \frac{4}{9} = 108.$$

Answer: (C)

31.

SAT CHALLENGE · CALCULATOR

Real numbers x and y satisfy

$$\sqrt[3]{2^x} \cdot 4^y = 256 \quad \text{and} \quad \frac{16^x}{\sqrt[3]{4^y}} = 2^{22}.$$

What is the value of xy ?

- Write everything as powers of 2: $\sqrt[3]{2^x} = 2^{x/3}$, $4^y = 2^{2y}$, $256 = 2^8$. First equation: $2^{x/3+2y} = 2^8$, so $\frac{x}{3} + 2y = 8$, i.e. $x + 6y = 24$.
- Second equation: $16^x = 2^{4x}$ and $\sqrt[3]{4^y} = 2^{2y/3}$, so $2^{4x-2y/3} = 2^{22}$, giving $4x - \frac{2y}{3} = 22$, i.e. $6x - y = 33$.
- From $6x - y = 33$: $y = 6x - 33$. Substitute: $x + 6(6x - 33) = 24 \Rightarrow 37x = 222 \Rightarrow x = 6$, so $y = 3$.
- Check: $\frac{6}{3} + 2(3) = 8$ and $4(6) - \frac{2(3)}{3} = 22$.
- $xy = 6 \cdot 3 = 18$.

Answer: 18**32.**

SAT CHALLENGE · NO CALCULATOR NEEDED

For a real number x , $4^x - 2^{x+2} - 32 = 0$. What is the value of 2^{x+1} ?

- (A) 16 ✓ (B) 32
- (C) -8 (D) 8

- Write $4^x = (2^x)^2$ and $2^{x+2} = 4 \cdot 2^x$. Let $u = 2^x$ (so $u > 0$): $u^2 - 4u - 32 = 0$.
- Factor: $(u - 8)(u + 4) = 0$, so $u = 8$ or $u = -4$.
- 2^x is always positive, so $u = -4$ is rejected. $u = 8$, i.e. $x = 3$.
- $2^{x+1} = 2u = 16$.

Answer: (A)

33.

SAT CHALLENGE · CALCULATOR

The size of a bacterial colony, in thousands of cells, is modeled by $V(t) = 500 \cdot 2^{t/2} \cdot 4^{t/3} \cdot 16^{-t/12}$, where t is the number of hours since the start of observation. The size of the colony is multiplied by a constant factor c every 6 hours. What is the value of c ?

1. Write every base as a power of 2: $4^{t/3} = 2^{2t/3}$ and $16^{-t/12} = 2^{-t/3}$.
2. Add the exponents: $\frac{t}{2} + \frac{2t}{3} - \frac{t}{3} = \frac{3t + 4t - 2t}{6} = \frac{5t}{6}$, so $V(t) = 500 \cdot 2^{5t/6}$.
3. Rewrite with the interval in the exponent: $2^{5t/6} = (2^5)^{t/6} = 32^{t/6}$.
4. $c = 32$. (Check: $V(t+6)/V(t) = 2^5 = 32$.)

Answer: 32**34.**

SAT CHALLENGE · CALCULATOR

The function f is defined by $f(x) = \frac{9^x \cdot 12^{x+1}}{8^x}$. Which of the following expressions is equivalent to $f(x)$ and displays the value of $f(2)$ as a coefficient?

(A) $2187 \left(\frac{27}{2}\right)^{x+2}$

(B) $162 \left(\frac{27}{2}\right)^{x-1}$

(C) $\frac{729}{4} \left(\frac{27}{2}\right)^{x-2}$

(D) $2187 \left(\frac{27}{2}\right)^{x-2} \checkmark$

1. Prime factors: $9^x = 3^{2x}$, $12^{x+1} = 2^{2x+2} \cdot 3^{x+1}$, $8^x = 2^{3x}$.
2. $f(x) = 3^{2x+x+1} \cdot 2^{2x+2-3x} = 3^{3x+1} \cdot 2^{2-x} = 12 \left(\frac{27}{2}\right)^x$.
3. $f(2) = 12 \cdot \frac{729}{4} = 2187$.
4. Shift the exponent: $f(x) = f(2) \left(\frac{27}{2}\right)^{x-2} = 2187 \left(\frac{27}{2}\right)^{x-2}$.
5. Other options: $162 \left(\frac{27}{2}\right)^{x-1}$ is equivalent but shows $f(1)$; $\frac{729}{4} \left(\frac{27}{2}\right)^{x-2}$ and the $x+2$ form are not equivalent.

Answer: (D)

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 2^x + k$, where k is a constant. For a real number p , $f(p) = 9$ and $f(-p) = \frac{21}{4}$. What is the value of p ?

1. Let $t = 2^p$, so $t > 0$ and $2^{-p} = \frac{1}{t}$. Then $t + k = 9$ and $\frac{1}{t} + k = \frac{21}{4}$.
2. Subtract to eliminate k : $t - \frac{1}{t} = \frac{15}{4} \Rightarrow 4t^2 - 15t - 4 = 0 \Rightarrow (4t + 1)(t - 4) = 0$.
3. $t = -\frac{1}{4}$ is rejected because $2^p > 0$; so $t = 4$.
4. $2^p = 4 \Rightarrow p = 2$. (Check: $k = 5$; $f(2) = 9$, $f(-2) = \frac{1}{4} + 5 = \frac{21}{4}$.)

Answer: 2**36.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = c \cdot b^{x+3}$, where c and b are positive constants. If $f(1) = 1296$ and $f(3) = 729$, then for each increase of 1 in the value of x , $f(x)$ decreases by $p\%$. What is the value of p ?

1. $f(1) = cb^4 = 1296$ and $f(3) = cb^6 = 729$.
2. Divide to cancel c : $b^2 = \frac{729}{1296} = \frac{9}{16}$. Since $b > 0$, $b = \frac{3}{4}$.
3. Each unit increase in x multiplies f by $\frac{3}{4}$, a decrease of $1 - \frac{3}{4} = \frac{1}{4} = 25\%$.
4. $p = 25$. (The value of c is not needed.)

Answer: 25

37.

SAT CHALLENGE · CALCULATOR

The value V , in dollars, of a delivery van x years after it is purchased is modeled by $V(x) = 18,000(0.49)^{x/2}$. Which of the following best describes how the van's value changes each year?

(A) It decreases by 51% each year. (B) It decreases by 24.5% each year.

(C) It decreases by 30% each year. ✓ (D) It decreases by 70% each year.

1. Rewrite the exponent: $(0.49)^{x/2} = \left((0.49)^{1/2}\right)^x = (0.7)^x$.
2. So $V(x) = 18,000(0.7)^x$ and each year the value is multiplied by 0.7.
3. The van keeps 70% of its value, so it loses $1 - 0.7 = 0.30$, i.e. 30%, each year.
4. Check: $\frac{V(3)}{V(2)} = \frac{(0.49)^{1.5}}{(0.49)^1} = 0.49^{0.5} = 0.7$. The option 51% is $1 - 0.49$ (the two-year loss).

Answer: (C)

38.

SAT CHALLENGE · CALCULATOR

The area covered by a coral colony increases by 33.1% every 3 years. If the colony grows by the same percent each year, by what percent does its area increase each year?

1. Over 3 years the area is multiplied by 1.331. If the yearly factor is b , then $b^3 = 1.331$.
2. Take the cube root: $b = \sqrt[3]{1.331} = 1.1$, because $1.1^3 = 1.331$.
3. A factor of 1.1 is an increase of $1.1 - 1 = 0.10$, that is 10% per year.

Answer: 10

39.

SAT CHALLENGE · CALCULATOR

The function f is defined by $f(x) = ab^x + c$, where a , b , and c are constants and $b > 1$. The values $f(2) = 14$, $f(4) = 20$, and $f(6) = 32$ are known. What is the value of c ?

1. The inputs 2, 4, 6 are equally spaced, so $f(2) - c$, $f(4) - c$, $f(6) - c$ form a geometric sequence: $(20 - c)^2 = (14 - c)(32 - c)$.
2. $400 - 40c + c^2 = 448 - 46c + c^2 \Rightarrow 6c = 48 \Rightarrow c = 8$.
3. Check: 6, 12, 24 has constant ratio $2 = b^2$, so $b = \sqrt{2}$ and $a = 3$; indeed $3 \cdot 2 + 8 = 14$, $3 \cdot 4 + 8 = 20$, $3 \cdot 8 + 8 = 32$.

Answer: 8**40.**

SAT CHALLENGE · NO CALCULATOR NEEDED

For all real values of x ,

$$\frac{6^{2x+3}}{2^{2x-1}} = k \cdot 9^x$$

where k is a constant. What is the value of k ?

1. Factor the base: $6^{2x+3} = 2^{2x+3} \cdot 3^{2x+3}$.
2. Divide by 2^{2x-1} : the powers of 2 give $2^{(2x+3)-(2x-1)} = 2^4 = 16$.
3. The powers of 3 give $3^{2x+3} = 3^3 \cdot 3^{2x} = 27 \cdot 9^x$.
4. So the left side equals $16 \cdot 27 \cdot 9^x = 432 \cdot 9^x$ and $k = 432$.

Answer: 432

41.

SAT CHALLENGE · CALCULATOR

A patient takes a 100-milligram dose of a medication at 8:00 a.m. and again at 8:00 p.m. every day, beginning at 8:00 a.m. on Monday. The dose is absorbed instantly, and between any two consecutive doses exactly half of the medication present in the body is eliminated. To the nearest tenth of a milligram, how much medication is in the body immediately after the 8:00 a.m. dose on Thursday?

(A) 196.9

(B) 198.4 ✓

(C) 199.2

(D) 187.5

- Count the doses: Mon a.m., Mon p.m., Tue a.m., Tue p.m., Wed a.m., Wed p.m., Thu a.m. That is 7 doses, with 6 twelve-hour intervals (two per day).
- Let a_n be the amount just after dose n : $a_1 = 100$ and $a_{n+1} = \frac{1}{2}a_n + 100$. Then $a_7 = 100 \left(1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^6}\right)$.
- Geometric sum: $100 \cdot \frac{1 - (1/2)^7}{1 - 1/2} = 200 \left(1 - \frac{1}{128}\right) = 200 \cdot \frac{127}{128} = 198.4375$.
- To the nearest tenth: 198.4 mg. (The options 196.9 and 199.2 correspond to 6 and 8 doses; 187.5 to 4 doses.)

Answer: (B)

42.

SAT CHALLENGE · CALCULATOR

The function $P(t) = \frac{120}{1 + 5(0.8)^t}$ models the number of active users of an app, in thousands, t weeks after launch, for $0 \leq t \leq 10$. Consider the following statements.

- P is increasing on the interval $0 \leq t \leq 10$.
- There is a time t in the interval at which $P(t) = 60$.
- There is a time t in the interval at which $P(t) = 80$.

Which of the statements must be true?

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II, and III

- I: as t increases, $(0.8)^t$ decreases, so the denominator $1 + 5(0.8)^t$ decreases and P increases. True.
- Since P is continuous and increasing, it attains exactly the values between $P(0) = \frac{120}{6} = 20$ and $P(10) = \frac{120}{1 + 5(0.8)^{10}} \approx \frac{120}{1.537} \approx 78.1$.
- II: 60 lies between 20 and 78.1. Directly: $1 + 5(0.8)^t = 2 \Rightarrow (0.8)^t = 0.2 \Rightarrow t = \frac{\ln 0.2}{\ln 0.8} \approx 7.2$, inside the interval. True.
- III: $80 > 78.1$. Directly: $1 + 5(0.8)^t = 1.5 \Rightarrow (0.8)^t = 0.1 \Rightarrow t = \frac{\ln 0.1}{\ln 0.8} \approx 10.3 > 10$. Outside the interval, so False (the bound of 120 is never reached, and on this interval the maximum is only ≈ 78.1).
- Only I and II must be true.

Answer: (B)

43.

SAT CHALLENGE · CALCULATOR

The function f is defined by $f(x) = a \cdot b^x$, where a and b are positive constants. For every value of x , $f(x + 2)$ is 44% greater than $f(x)$. For every value of x , $f(x + 3)$ is $p\%$ greater than $f(x)$. What is the value of p ?

(A) 132

(B) 60

(C) 66

(D) 72.8 ✓

1. $f(x + 2) = 1.44 f(x)$, so $\frac{a b^{x+2}}{a b^x} = b^2 = 1.44$.

2. Hence $b = 1.2$ (since $b > 0$).

3. $f(x + 3) = b^3 f(x) = (1.2)^3 f(x) = 1.728 f(x)$.

4. 1.728 is 72.8% greater than 1, so $p = 72.8$. (Option 66 = $1.5 \cdot 44$ scales the percent linearly; $60 = 3 \cdot 20$ ignores compounding; $132 = 3 \cdot 44$.)

Answer: (D)

44.

SAT CHALLENGE · NO CALCULATOR NEEDED

The value V , in dollars, of a piece of lab equipment t years after it was purchased is modeled by

$$V(t) = 2400(0.64)^{t/2.5}.$$

Which of the following is the best interpretation of the model?

(A) It falls 36% every 2.5 years. ✓

(B) It falls 36% every 5 years.

(C) It falls 64% every 2.5 years.

(D) It falls 2.5% every year.

1. The exponent is $\frac{t}{2.5}$, so it increases by 1 each time t increases by 2.5 years.

2. Each increase of 1 in the exponent multiplies V by 0.64. So every 2.5 years the value is multiplied by 0.64.

3. A factor of 0.64 is a decrease of $1 - 0.64 = 0.36 = 36\%$.

4. So the value decreases by 36% every 2.5 years.

5. (Over 5 years the factor is $0.64^2 \approx 0.41$, a 59% drop, not 36%.)

Answer: (A)

45.

For all real numbers x ,

$$8^x \cdot 4^{x/3} \cdot (\sqrt{2})^x = b^x,$$

where b is a positive constant. Which of the following is equal to 2?

(A) $b^{-6/25}$

(B) $b^{6/25}$ ✓

(C) $b^{25/6}$

(D) b^{25}

1. Write each factor as a power of 2: $8^x = 2^{3x}$, $4^{x/3} = 2^{2x/3}$, $(\sqrt{2})^x = 2^{x/2}$.2. Multiply (add exponents): $2^{x(3+\frac{2}{3}+\frac{1}{2})} = 2^{\frac{25}{6}x}$.3. So $b^x = \left(2^{25/6}\right)^x$ for all x , which gives $b = 2^{25/6}$.4. Raise both sides to the reciprocal power $\frac{6}{25}$: $b^{6/25} = 2$.5. (Evaluating at $x = 1$ gives b directly: $8 \cdot 4^{1/3} \cdot \sqrt{2}$.)**Answer: (B)**

46.

The function f is defined for $x \geq 0$ by

$$f(x) = 48 \left(\frac{1}{8}\right)^{\frac{2-x}{3}}.$$

Which of the following is equivalent to $f(x)$ for all $x \geq 0$ and displays the value of $f(0)$ as a constant factor?

(A) $f(x) = 12 \cdot \left(\frac{1}{2}\right)^x$

(B) $f(x) = 192 \cdot \left(\frac{1}{2}\right)^x$

(C) $f(x) = 12 \cdot 2^x$ ✓

(D) $f(x) = 48 \cdot 2^x$

1. Write $\frac{1}{8} = 2^{-3}$. Then $\left(\frac{1}{8}\right)^{(2-x)/3} = 2^{-3 \cdot (2-x)/3} = 2^{-(2-x)} = 2^{x-2}$.2. So $f(x) = 48 \cdot 2^{x-2} = 48 \cdot \frac{2^x}{4} = 12 \cdot 2^x$.3. Check: $f(0) = 48 \left(\frac{1}{8}\right)^{2/3} = 48 \cdot \frac{1}{4} = 12$, which is the constant factor, and $f(1) = 48 \left(\frac{1}{8}\right)^{1/3} = 24 = 12 \cdot 2^1$.4. The negative exponent $\frac{2-x}{3}$ reverses the effect of the fractional base, so f grows. Only $12 \cdot 2^x$ matches.**Answer: (C)**

47.

SAT CHALLENGE · NO CALCULATOR NEEDED

The number of bacteria in a colony t years after January 1, 2020 is modeled by $A(t) = 300 \cdot 4^{t/3}$. Which of the following gives the number of bacteria m months after January 1, 2023?

(A) $1200 \cdot 2^{m/18}$ ✓

(B) $300 \cdot 2^{m/18}$

(C) $1200 \cdot 2^{m/36}$

(D) $900 \cdot 2^{m/18}$

1. January 1, 2023 is 3 years after January 1, 2020, so $t = 3 + \frac{m}{12}$.
2. $A = 300 \cdot 4^{(3 + \frac{m}{12})/3} = 300 \cdot 4^{1 + \frac{m}{36}} = 300 \cdot 4 \cdot 4^{m/36} = 1200 \cdot 4^{m/36}$.
3. Change the base: $4^{m/36} = (2^2)^{m/36} = 2^{m/18}$. So $A = 1200 \cdot 2^{m/18}$.
4. Check: $m = 0$ gives $1200 = A(3) = 300 \cdot 4$. $m = 36$ gives $1200 \cdot 4 = 4800 = A(6) = 300 \cdot 16$.

Answer: (A)

48.

SAT CHALLENGE · NO CALCULATOR NEEDED

Let $g(x) = 7 + 9 \left(\frac{1}{4}\right)^x$ for all real numbers x . Consider the following statements.

- I. $g(x) > 7$ for every real number x .
- II. The minimum value of g is 7.
- III. g is a decreasing function.

Which of the statements must be true?

(A) **I and III only** ✓

(B) I and II only

(C) II and III only

(D) I, II, and III

1. $\left(\frac{1}{4}\right)^x > 0$ for every x , so $9 \left(\frac{1}{4}\right)^x > 0$ and $g(x) > 7$. Statement I is true.
2. As x grows, $\left(\frac{1}{4}\right)^x \rightarrow 0$, so $g(x) \rightarrow 7$, but the term is never 0, so $g(x) = 7$ is never attained. There is no minimum value; 7 is only a lower bound. Statement II is false.
3. The base $\frac{1}{4}$ is between 0 and 1, so $\left(\frac{1}{4}\right)^x$ decreases as x increases; multiplying by $9 > 0$ and adding 7 keeps that. Statement III is true.
4. Only I and III are true.

Answer: (A)